Open Access Article. Published on 03 November 2023. Downloaded on 3/24/2026 11:07:18 PM.
This article is licensed under a Creative Commons Attribution 3.0 Unported Licence.

25 D. Aurbach, I. Weissman, A. Zaban and P. Dan, *Electrochim. Acta*, 1999, **45**, 1135.
26 U. Langklotz, M. Schneider and A. Michaelis, *J. Ceram. Sci. Technol.*, 2013, **4**(2), 69–76.
27 T. Osaka, T. Momma, T. Tajima and Y. Matsumoto, *J. Electrochem. Soc.*, 1995, **142**, 1057.
28 L.-Q. Zheng, S.-J. Li, H.-J. Lin, Y.-Y. Miao, L. Zhu and Z.-J. Zhang, *Russ. J. Electrochem.*, 2014, **50**, 904.
29 R. M. Kasse, N. R. Geise, J. S. Ko, J. Nelson Weker, H.-G. Steinrück and M. F. Toney, *J. Mater. Chem. A*, 2020, **8**, 16960.
30 W. Weber, V. Kraft, M. Grützke, R. Wagner, M. Winter and S. Nowak, *J. Chromatogr. A*, 2015, **1394**, 128.
31 M. Stich, M. Göttlinger, M. Kurniawan, U. Schmidt and A. Bund, *J. Phys. Chem. C*, 2018, **122**, 8836.
32 F. Huttner, W. Haselrieder and A. Kwade, *Energy Technol.*, 2020, **8**, 1900245.
33 H. Maleki, G. Deng, A. Anani and J. Howard, *J. Electrochem. Soc.*, 1999, **146**, 3224.
34 S. Solchenbach, M. Metzger, M. Egawa, H. Beyer and H. A. Gasteiger, *J. Electrochem. Soc.*, 2018, **165**(13), A3022–A3028.
35 A. T. Freiberg, J. Sicklinger, S. Solchenbach and H. A. Gasteiger, *Electrochim. Acta*, 2020, **346**, 136271.
36 R. Spotnitz and J. Franklin, *J. Power Sources*, 2003, **113**, 81.
37 W. D. Machin and F. C. Tompkins, *Trans. Faraday Soc.*, 1966, **62**, 2205–2218.
38 H. Shin, J. Park, S. Han, A. M. Sastry and W. Lu, *J. Power Sources*, 2015, **277**, 169.
39 D. Bedrov, O. Borodin and J. B. Hooper, *J. Phys. Chem. C*, 2017, **121**, 16098.
40 H. Ye, S. Gui, Z. Wang, J. Chen, Q. Liu, X. Zhang, P. Jia, Y. Tang, T. Yang, C. Du, L. Geng, H. Li, Q. Dai, Y. Tang, L. Zhang, H. Yang and J. Huang, *ACS Appl. Mater. Interfaces*, 2021, **13**, 44479.
41 C. Stetson, Y. Yin, C.-S. Jiang, S. C. DeCaluwe, M. Al-Jassim, N. R. Neale, C. Ban and A. Burrell, *ACS Energy Lett.*, 2019, **4**, 2770.
42 P. Atkins, and J. de Paula, *Thermodynamics and Kinetics*, W.H. Freeman and Co, New York, 2005.
43 H. Maleki and J. N. Howard, *J. Power Sources*, 2004, **137**, 117.
44 *Renata Batteries Datasheet*, 2019, ICP622540PMT.
45 A. M. Colclasure and R. J. Kee, *Electrochim. Acta*, 2010, **55**, 8960.
46 B. Rieger, S. Schlueter, S. V. Erhard, J. Schmalz, G. Reinhart and A. Jossen, *J. Energy Storage*, 2016, **6**, 213.
47 Y. Wang, S. Nakamura, M. Ue and P. B. Balbuena, *J. Am. Chem. Soc.*, 2001, **123**, 11708.
48 *MATLAB, Version 9.12.0.1956245 (R2022a)*, The MathWorks Inc, Natick, Massachusetts, 2022.
49 G. V. Zhuang, K. Xu, H. Yang, T. R. Jow and P. N. Ross, *J. Phys. Chem. B*, 2005, **109**, 17567.
50 L. Wang, A. Menakath, F. Han, Y. Wang, P. Y. Zavalij, K. J. Gaskell, O. Borodin, D. Iuga, S. P. Brown, C. Wang, K. Xu and B. W. Eichhorn, *Nat. Chem.*, 2019, **11**, 789.
51 X. Xie, E. W. Clark Spotte-Smith, M. Wen, H. D. Patel, S. M. Blau and K. A. Persson, *J. Am. Chem. Soc.*, 2021, **143**, 13245.
52 M. Nie, D. Chalasani, D. P. Abraham, Y. Chen, A. Bose and B. L. Lucht, *J. Phys. Chem. C*, 2013, **117**, 1257.
53 F. A. Soto, Y. Ma, J. M. La Martinez de Hoz, J. M. Seminario and P. B. Balbuena, *Chem. Mater.*, 2015, **27**, 7990.
54 B. Lu, D. Cheng, B. Sreenarayanan, W. Li, B. Bhamwala, W. Bao and Y. S. Meng, *ACS Energy Lett.*, 2023, **8**, 3230.
55 B. S. Vishnugopi, M. T. Hasan, H. Zhou and P. P. Mukherjee, *ACS Energy Lett.*, 2023, **8**, 398.
56 P. Röder, B. Stiaszny, J. C. Ziegler, N. Baba, P. Lagaly and H.-D. Wiemhöfer, *J. Power Sources*, 2014, **268**, 315.
57 C. Jayawardana, N. Rodrigo, B. Parimalam and B. L. Lucht, *ACS Energy Lett.*, 2021, **6**, 3788.

 © 2023 The Author(s). Published by the Royal Society of Chemistry